

# Final Migration Report

Governed sandbox migration report

## 1. Executive Summary

| Field                     | Value                                        |
|---------------------------|----------------------------------------------|
| Run ID                    | v2-604c4a0e-s4                               |
| Generated At              | 2026-06-18T15:07:41Z                         |
| Legacy Baseline           | Spring Boot 2.1.6.RELEASE / Java 11          |
| Current Application State | Spring Boot 4.0 / Java 21                    |
| Latest Completed Stage    | Stage 4: springboot-3.5-java21-to-4.0-java21 |
| Migration Duration        | 1904.971s                                    |
| Approval Decision         | approved                                     |
| Transform Status          | TRANSFORM_APPLIED_IN_SANDBOX                 |
| Build Status              | BUILD_PASSED_IN_SANDBOX                      |
| Test Status               | PASS_WITH_WARNINGS                           |
| Proof Level               | compiled                                     |
| Dependency Policy         | PASS                                         |

This report describes the full migration journey from Spring Boot 2.1.6.RELEASE / Java 11 to Spring Boot 4.0 / Java 21. The migration was executed as a controlled sandbox modernization flow with human approval before transformation.

Migration completed for the full path Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 4.0 / Java 21 with approval decision approved, transform status TRANSFORM\_APPLIED\_IN\_SANDBOX, build status BUILD\_PASSED\_IN\_SANDBOX, and test status PASS\_WITH\_WARNINGS. Elapsed duration: 1904.971s. Primary change summary: Full migration path: Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 4.0 / Java 21

## 2. Migration Story

The application started from Spring Boot 2.1.6.RELEASE / Java 11 and progressed through the staged migration pipeline until it reached Spring Boot 4.0 / Java 21. The latest completed stage was Stage 4: springboot-3.5-java21-to-4.0-java21, where the final transition was Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21.

Migration flow followed during this run:

Analyze -> Plan -> Assess -> Human Approval -> Sandbox Transform -> Build Validation -> Test Validation -> Final Report

## Migration Process

- \* Legacy application baseline: Spring Boot 2.1.6.RELEASE / Java 11
- \* Latest completed stage: Stage 4: springboot-3.5-java21-to-4.0-java21
- \* Completed stage transition: Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21
- \* Current application state: Spring Boot 4.0 / Java 21
- \* Final executed target: Spring Boot 4.0 / Java 21
- \* Human approval decision: approved
- \* Sandbox transform result: TRANSFORM\_APPLIED\_IN\_SANDBOX
- \* Build result: BUILD\_PASSED\_IN\_SANDBOX
- \* Test result: PASS\_WITH\_WARNINGS
- \* Proof level achieved: compiled
- \* Repair loop outcome: NOT\_IMPLEMENTED
- \* Dependency policy outcome: PASS

## 3. Current Technical State

| Area                          | Value                               |
|-------------------------------|-------------------------------------|
| Legacy application baseline   | Spring Boot 2.1.6.RELEASE / Java 11 |
| Latest completed stage source | Spring Boot 3.5 / Java 21           |
| Current application state     | Spring Boot 4.0 / Java 21           |
| Spring Framework target       | 7.0.x                               |
| Risk Level                    | high                                |
| Strategy                      | staged_openrewrite_sandbox          |
| Fallback Profile              | not captured                        |
| Production Allowed            | false                               |

## 4. Phase Status

| Phase             | Status                       | Explanation                                                              |
|-------------------|------------------------------|--------------------------------------------------------------------------|
| Human Approval    | approved                     | Human approval was required before sandbox transformation.               |
| Sandbox Transform | TRANSFORM_APPLIED_IN_SANDBOX | Code changes were applied only inside the sandbox workspace.             |
| Build Validation  | BUILD_PASSED_IN_SANDBOX      | The migrated sandbox candidate was validated by build execution.         |
| Test Validation   | PASS_WITH_WARNINGS           | Existing tests were run against the migrated sandbox candidate.          |
| Proof Level       | compiled                     | This is the highest deterministic proof level reached during validation. |
| Repair Loop       | NOT_IMPLEMENTED              | Indicates whether repair logic was needed to stabilize the migration.    |
| Dependency Policy | PASS                         | Shows whether dependency policy checks passed or raised review items.    |

## 5. Stage-By-Stage Journey

The migration was not a single jump. It was executed as a staged progression so each version boundary could be validated more safely.

| Stage   | Profile                             | Transition                                                       | Status                       | Duration |
|---------|-------------------------------------|------------------------------------------------------------------|------------------------------|----------|
| Stage 1 | springboot-2.1.6-to-2.7-java11      | Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 2.7 / Java 11 | TRANSFORM_APPLIED_IN_SANDBOX | 56m13s   |
| Stage 2 | springboot-2.7-to-3.5-java17        | Spring Boot 2.7 / Java 11 -> Spring Boot 3.5 / Java 17           | TRANSFORM_APPLIED_IN_SANDBOX | 56m13s   |
| Stage 3 | springboot-3.5-java17-to-java21     | Spring Boot 3.5 / Java 17 -> Spring Boot 3.5 / Java 21           | TRANSFORM_APPLIED_IN_SANDBOX | 34m19s   |
| Stage 4 | springboot-3.5-java21-to-4.0-java21 | Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21           | TRANSFORM_APPLIED_IN_SANDBOX | 49m18s   |

Narrative highlights:

- \* Stage 1 moved the application from Spring Boot 2.1.6.RELEASE / Java 11 to Spring Boot 2.7 / Java 11, with overall stage status TRANSFORM\_APPLIED\_IN\_SANDBOX.
  - \* Stage 2 moved the application from Spring Boot 2.7 / Java 11 to Spring Boot 3.5 / Java 17, with overall stage status TRANSFORM\_APPLIED\_IN\_SANDBOX.
  - \* Stage 3 moved the application from Spring Boot 3.5 / Java 17 to Spring Boot 3.5 / Java 21, with overall stage status TRANSFORM\_APPLIED\_IN\_SANDBOX.
  - \* Stage 4 moved the application from Spring Boot 3.5 / Java 21 to Spring Boot 4.0 / Java 21, with overall stage status TRANSFORM\_APPLIED\_IN\_SANDBOX.
- 

## 6. What Changed

---

### What Changed

The most important migration changes recorded for this run are listed below.

- \* Full migration path: Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 4.0 / Java 21
  - \* Java remained at 21.
  - \* Spring Boot changed from 3.5 to 4.0.
  - \* Spring Framework changed from not\_captured to 7.0.x.
  - \* Stage 1 (springboot-2.1.6-to-2.7-java11): Spring Boot 2.1.6.RELEASE / Java 11 -> Spring Boot 2.7 / Java 11
  - \* Stage 2 (springboot-2.7-to-3.5-java17): Spring Boot 2.7 / Java 11 -> Spring Boot 3.5 / Java 17
  - \* Stage 3 (springboot-3.5-java17-to-java21): Spring Boot 3.5 / Java 17 -> Spring Boot 3.5 / Java 21
  - \* Stage 4 (springboot-3.5-java21-to-4.0-java21): Spring Boot 3.5 / Java 21 -> Spring Boot 4.0 / Java 21
  - \* Executed OpenRewrite recipes: org.openrewrite.java.spring.boot4.UpgradeSpringBoot\_4\_0.
- 

## 7. Validation Outcome

Validated areas:

- \* sandbox transform applied
- \* Maven build passed

Not validated by this sandbox run:

- \* SQL Server production behavior
- \* production DB scripts
- \* endpoint/business behavior
- \* production secrets/JWT/keystore validity
- \* deployment
- \* PR creation/merge

This means the migration is well described technically, but any production-readiness conclusion still requires human review of runtime behavior, infrastructure compatibility, and environment-specific risks.

---

## 8. Spring Boot 4 Notes

These are the main points that deserve attention for the final Boot 4 state:

- \* Spring Boot 4 target requires Spring Framework 7.x.
  - \* Spring Boot 4 target carries Jakarta EE 11 / Servlet 6.1 baseline risk.
  - \* Boot 3 deprecated APIs removed in Boot 4 must be reviewed.
  - \* Spring Cloud compatibility must be reviewed before sandbox execution.
  - \* Spring Security, Spring Data, Hibernate, and custom starter compatibility risk requires human review.
  - \* javax.\* leftovers must be treated as blockers for Boot 4 readiness.
  - \* Maven version and Java runtime must match Boot 4 / Java 21 validation gates.
  - \* Official Spring Boot guidance prefers upgrading to the latest 3.5.x before Boot 4; direct migration is sandbox-only and should fall back if unstable.
  - \* OPENREWRITE\_IMPACT\_HIGH: OpenRewrite impact is high; manual review is required before execution.
  - \* OPENREWRITE\_HIGH\_RISK\_FILES: OpenRewrite reported high-risk files: 42.
  - \* Spring Framework 7.x is required for Spring Boot 4.
  - \* Jakarta EE 11 / Servlet 6.1 baseline applies.
  - \* Spring Cloud compatibility must be reviewed.
  - \* Spring Security, Spring Data, Hibernate, and custom starter risk requires human review.
  - \* javax.\* leftovers must be eliminated.
  - \* Maven >= 3.6.3 and Java 21 runtime validation are required for this sandbox profile.
  - \* Official Boot guidance prefers latest 3.5.x before Boot 4; direct migration must use the fallback profile if unstable.
- 

## 9. Repair And Stabilization

| Field                 | Value           |
|-----------------------|-----------------|
| Enabled               | false           |
| Max Attempts          | 3               |
| Attempts Used         | 0               |
| Final Status          | NOT_IMPLEMENTED |
| Ledger                |                 |
| Copilot Used          | false           |
| Safe Patch Applied    | false           |
| Human Review Required | false           |

## 11. Dependency Policy Review

| Field                | Value                                                                 |
|----------------------|-----------------------------------------------------------------------|
| Status               | PASS                                                                  |
| Risks                | 0                                                                     |
| Blockers             | 0                                                                     |
| Copilot Advisory     | SKIPPED                                                               |
| Policy Patch Applied | false                                                                 |
| Report               | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\lv2-60 |

## 12. Timing

### Timing

| Field          | Value                                                                 |
|----------------|-----------------------------------------------------------------------|
| Total duration | 1904.971s                                                             |
| Timing report  | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\lv2-60 |
| Timing summary | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration\runs\lv2-60 |

## 13. Related Artifacts

| Artifact                      | Path                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------|
| approval_decision             | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_approval_decision             |
| approved_plan_lock            | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_approved_plan_lock            |
| transformation_execution_plan | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_transformation_execution_plan |
| migration_ledger              | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_migration_ledger              |
| post_transform_test_report    | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_post_transform_test_report    |
| orchestration_summary         | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_orchestration_summary         |
| timing_report                 | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_timing_report                 |
| timing_summary                | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_timing_summary                |
| final_migration_report        | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_final_migration_report        |
| final_migration_summary       | C:\Users\hamdaoui.ali\Downloads\MSA-COMMON-STG1\migration_final_migration_summary       |

## 15. Final Note

This document is intended to help a reviewer understand what happened during the migration, why the application is now in its current state, and which areas still require manual judgment. The deterministic run artifacts remain the source of truth.

POC-ready sandbox migration artifacts are captured under this run directory.